

ICT702 Data Visualisation

Explaining Visually to Influence with Data

Module 7



I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



Part 3

Choosing the Correct Chart to Tell Your Story

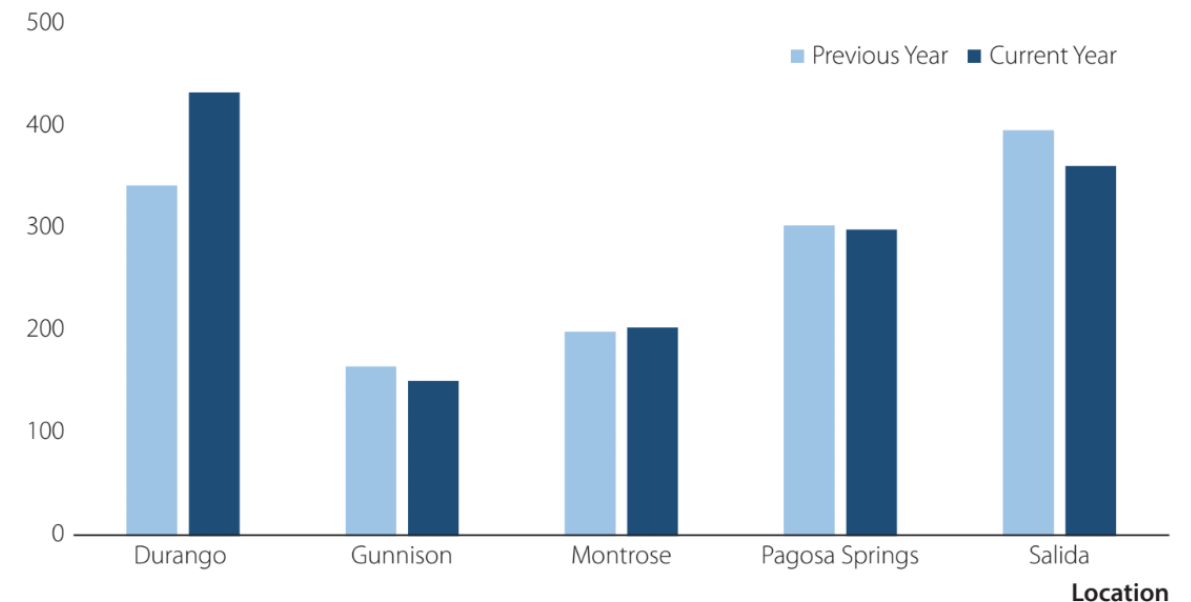
Data collected to Analyze Student Enrollment at Opening Horizon Locations

	A	B	C
1		Enrollments	
2	Location	Previous Year	Current Year
3	Durango	341	432
4	Gunnison	164	150
5	Montrose	198	202
6	Pagosa Springs	302	298
7	Salida	395	360

Insight that Durango and Salida enrollment has moved in different directions compared to last year not apparent to audience.

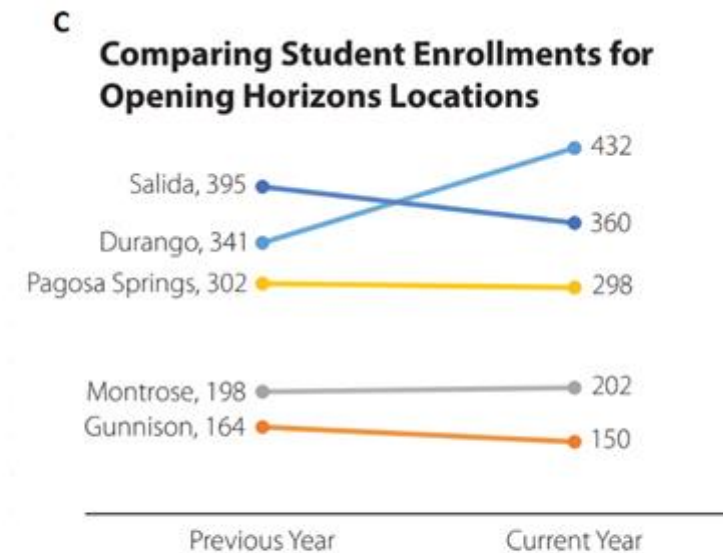
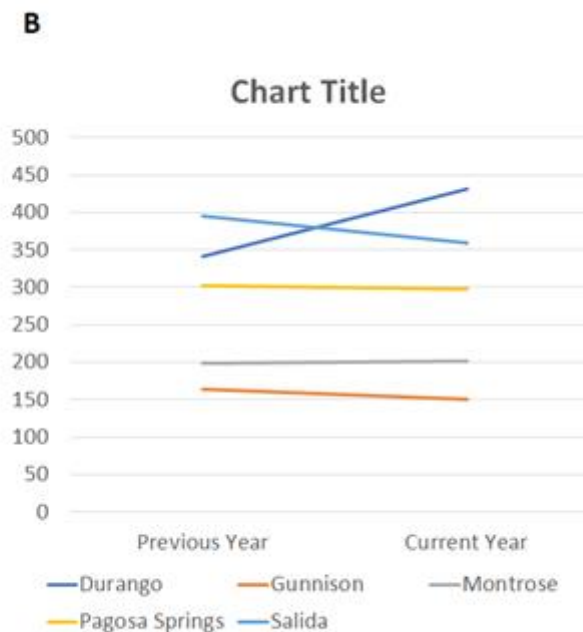
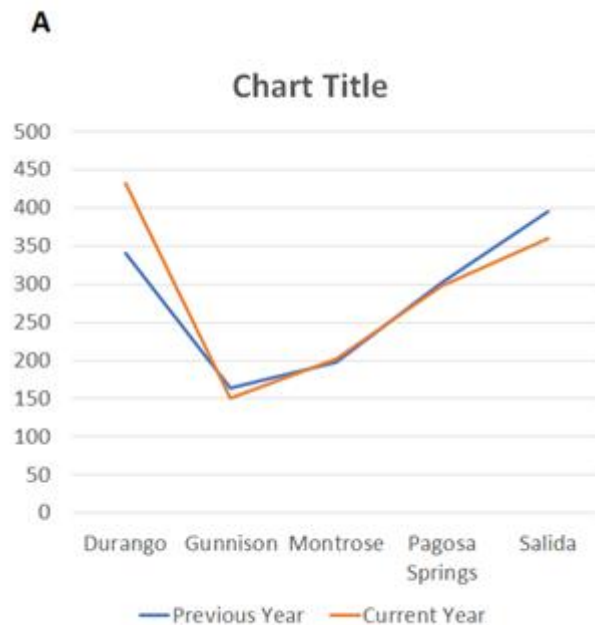
A clustered column chart for student enrollment data at Opening Horizon

Comparing Student Enrollments for Opening Horizons Locations
Number of Students at Start of Year



Build a Slope Chart

A slope chart for the student enrollment data at Opening Horizon

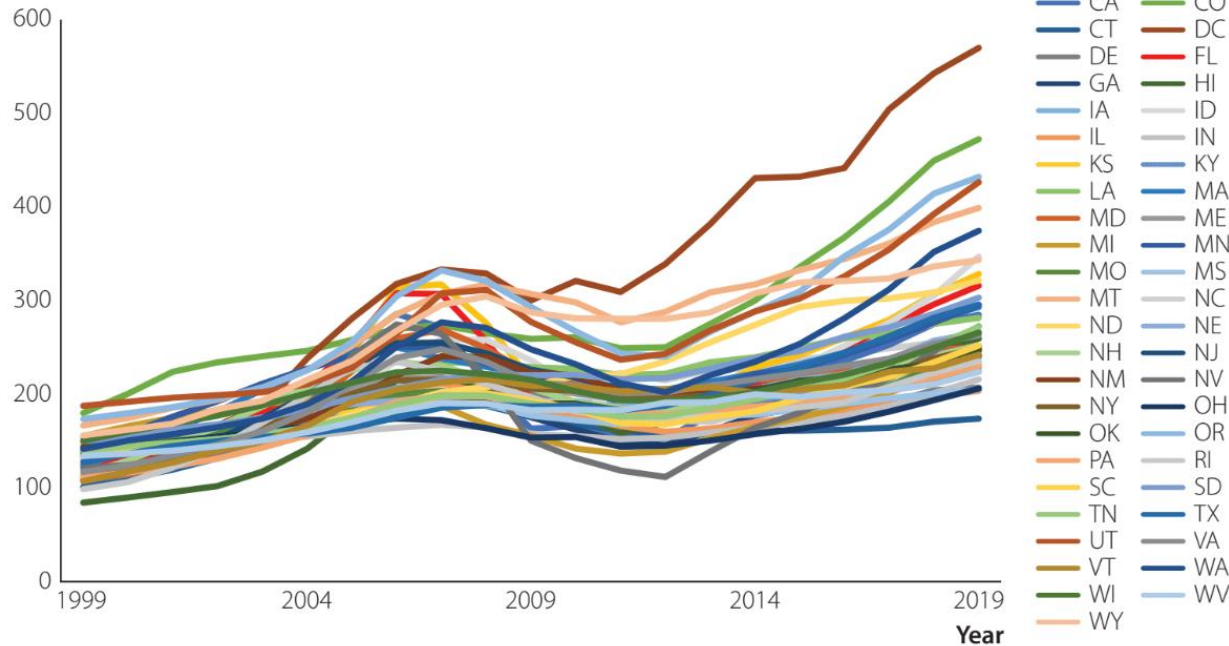


To create a slope chart in Excel, (A) build a simple line chart, (B) switch rows and columns, and (C) format the chart to its final version.

Using the Colour Attribute to Tell Your Story

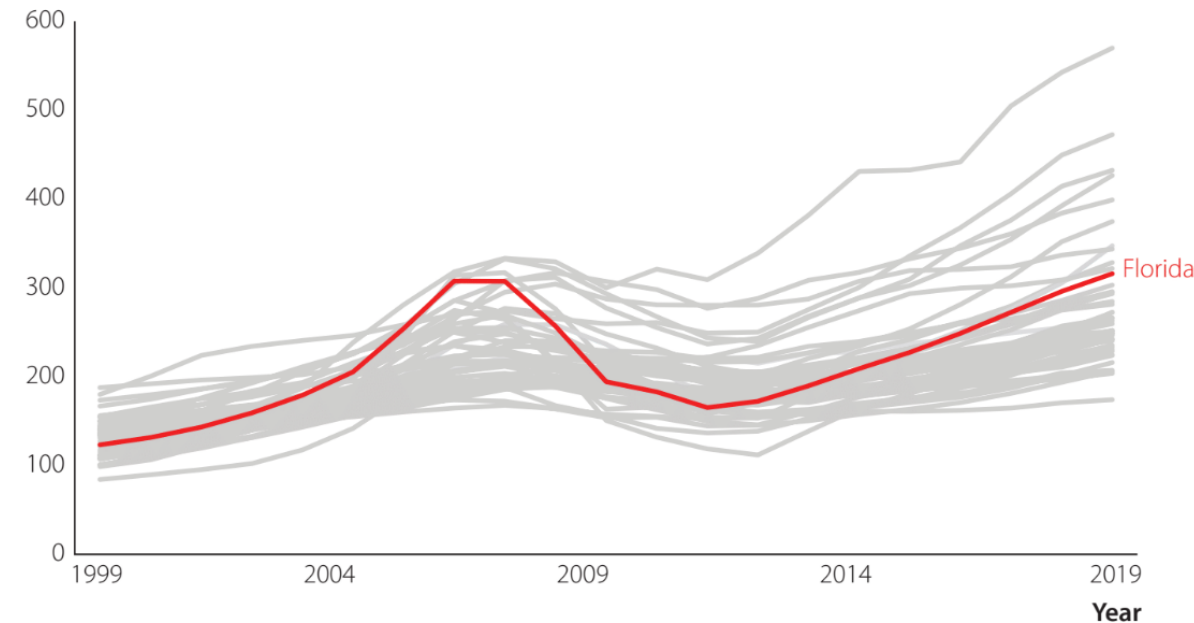
Quarterly House-Price Index by State

House-Price Index (Base = 1991)



Florida Experienced Extreme Boom and Bust During 2008 Subprime Mortgage Crisis

House-Price Index (Base = 1999)



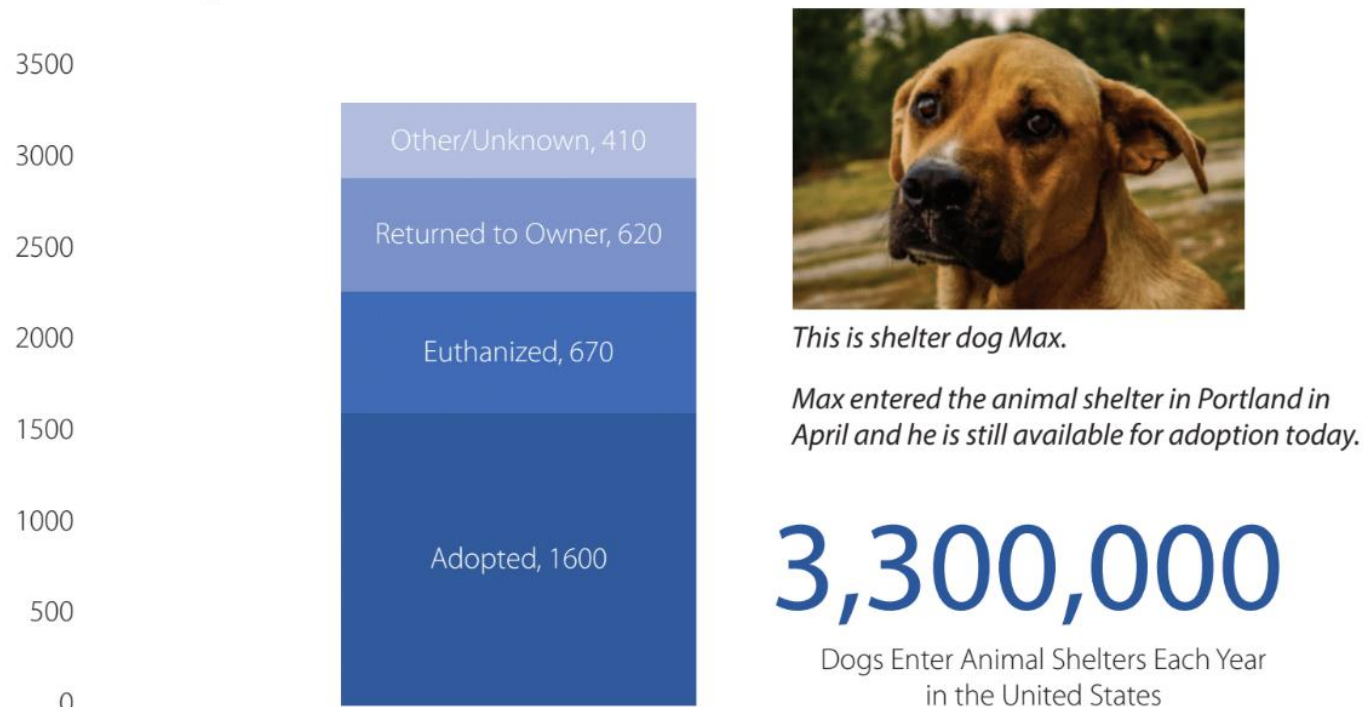
The preattentive attribute of colour emphasizes the impact the 2008 subprime mortgage crisis had on housing prices in Florida.

Using the Size Attribute to Tell Your Story

Incorporating the use of a BAN to emphasize the number of dogs that enter animal shelters each year

Fates of the Dogs Who Enter Animal Shelters Each Year

Number of Dogs (in 1000s)

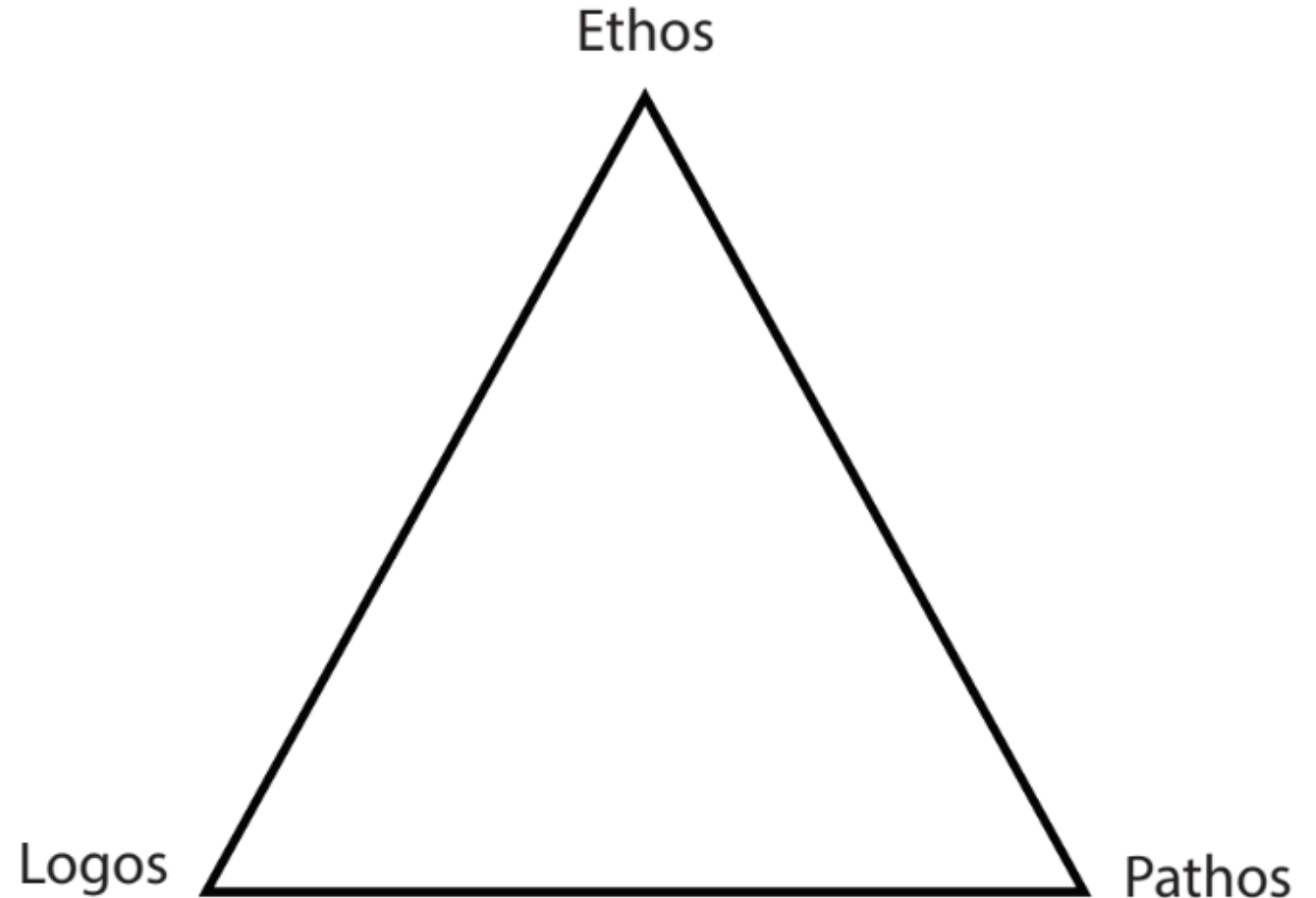


One simple but effective use of size as a preattentive attribute in data visualisation is using a **big associated number (BAN)**.

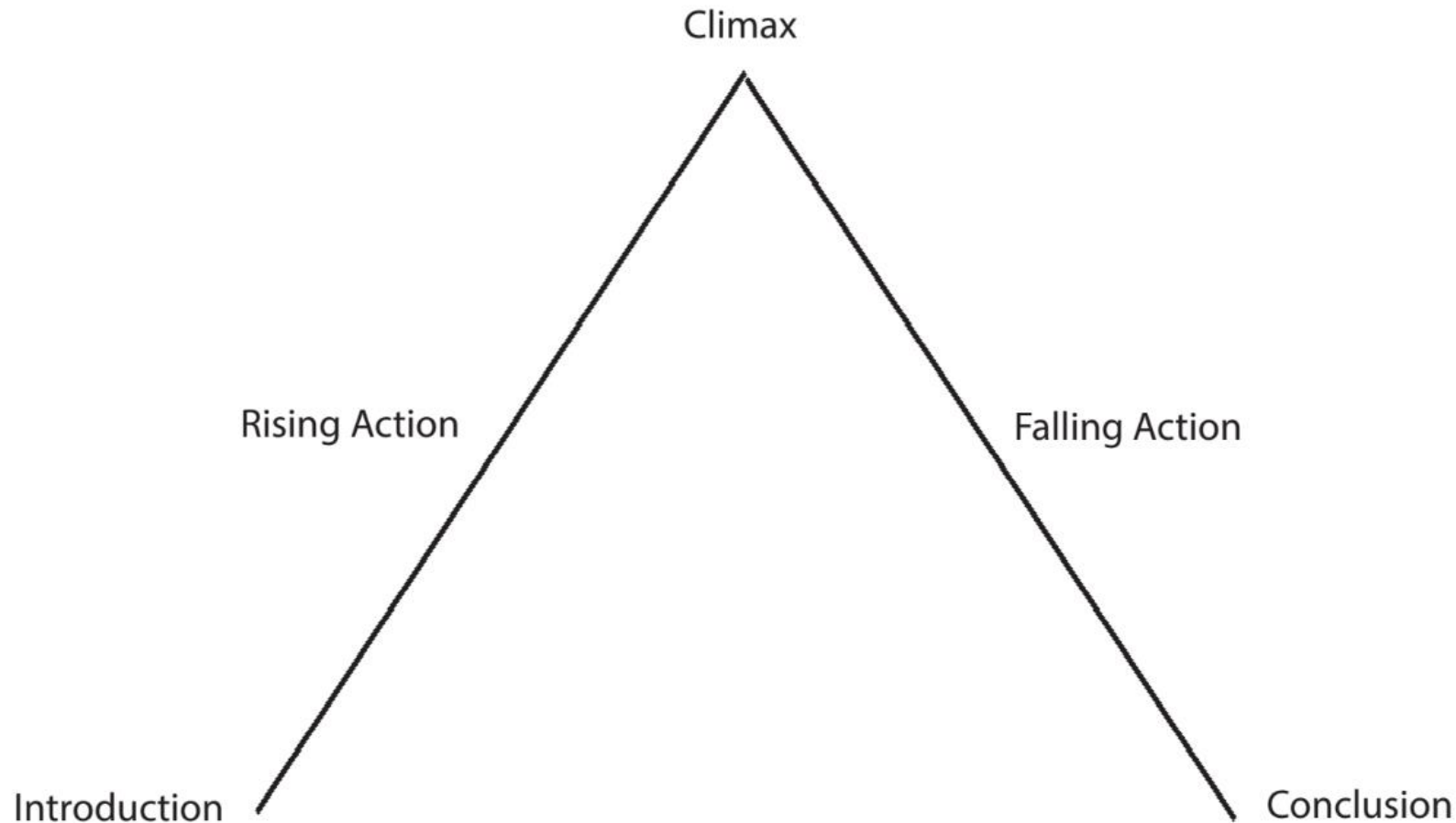
Aristotle's Rhetorical Triangle

The Rhetorical Triangle describes the three ways in which a compelling story should connect with an audience:

- Ethos
- Logos
- Pathos



Freytag's Pyramid



A **Freytag's Pyramid** divides a story into five distinct elements:

- 1. Introduction**
- 2. Rising action**
- 3. Climax**
- 4. Falling action**
- 5. Conclusion**

Application of Freytag's Pyramid to Storytelling

Hawaiian Bell presentation to Senior Team about customer service issues

Presentation mistakenly made following chronological order:

1. Detailed explanation of adopted data cleaning efforts
2. Alternative explanation of exploratory data analysis, including descriptive statistics
3. Aggregated findings from data

Issue: approach usually not best in terms of telling an effective story.

Application of Freytag's Pyramid to Storytelling

The Freytag's Pyramid provides better story to communicate insights

- **Introduction** includes basics of problem and why it is essential, by introducing a fictional protagonist, Liann
- **Rising action** describes Liann's struggle with customer service
- **Climax** develops conflict between Liann and the company
- **Falling action** uses narrative to provide recommendations on improving customer service.
- **Conclusion** outlines costs and benefits of recommendations and points out important assumptions/known limitations

Storyboarding

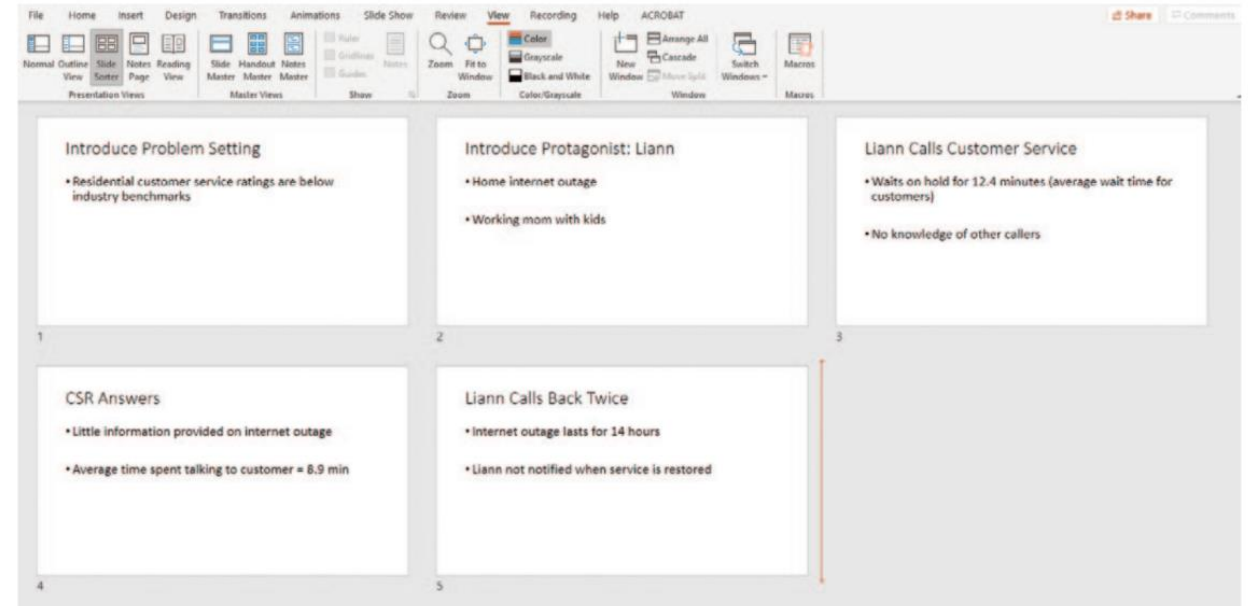
Partial sticky note storyboard for the Hawaiian Bell presentation



A partial storyboard using five yellow sticky notes. The notes are arranged in two rows. The top row contains three notes, and the bottom row contains two notes. Each note contains a title and a list of bullet points.

- INTRODUCE PROBLEM SETTING**
 - RESIDENTIAL CUSTOMER SERVICE RATINGS BELOW INDUSTRY STANDARDS
- INTRODUCE PROTAGONIST LIANN**
 - HOME INTERNET OUTAGE
 - WORKING MOM W/KIDS
- LIANN CALLS CUSTOMER SERVICE**
 - WAITS ON HOLD 12.4 MIN
 - NO KNOWLEDGE OF OTHER CALLERS
- CSR ANSWERS**
 - LITTLE INFO ON OUTAGE
 - AVERAGE TIME TALKING TO CUSTOMER = 8.9 MIN
- LIANN CALLS BACK TWICE**
 - INTERNET OUTAGE = 14 HRS
 - NOT NOTIFIED WHEN SERVICE RESTORED

Partial PowerPoint storyboard for the Hawaiian Bell presentation



A partial PowerPoint storyboard showing five slides in a grid. The slides are numbered 1 through 5. Each slide contains a title and a list of bullet points.

- 1. Introduce Problem Setting**
 - Residential customer service ratings are below industry benchmarks
- 2. Introduce Protagonist: Liann**
 - Home internet outage
 - Working mom with kids
- 3. Liann Calls Customer Service**
 - Waits on hold for 12.4 minutes (average wait time for customers)
 - No knowledge of other callers
- 4. CSR Answers**
 - Little information provided on internet outage
 - Average time spent talking to customer = 8.9 min
- 5. Liann Calls Back Twice**
 - Internet outage lasts for 14 hours
 - Liann not notified when service is restored

A simple visual organization of the main points of the story

Summary

In Module 7 you have learnt how:

- The characteristics of the audience may affect which types of data visualisations are most effective.
- To explain the importance of being able to empathize with data so that you may create the most effective visualisations and presentations.
- To use pre-attentive attributes such as colour and size to highlight particular insights from data.
- To build several new types of data visualisations such as the dot matrix chart and the slope chart.
- To illustrate the importance of storytelling and provide suggestions for structuring a presentation using concepts such as the Rhetorical Triangle and Freytag's Pyramid.
- To create storyboards using either sticky notes or a presentation software such as PowerPoint.

End of Part 3

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